Self-Contained Solar-Powered Inductive Power Transfer System for Wireless Electric Vehicle Charging

Zhicong Huang[1], Io Wa Iam[1,2,3], Iok U Hoi[1,2,3], Chi-Seng Lam[1,2,3], Pui-In Mak[1,2,3] and Rui P. Martins[1,2,3,4]
[1]*State Key Laboratory of Analog and Mixed-Signal VLSI, University of Macau,* Macao, China
[2]*Institute of Microelectronics, University of Macau,* Macao, China
[3]*Department of Electrical and Computer Engineering, Faculty of Science and Technology, University of Macau,* Macao, China
[4]on leave from *the Instituto Superior Técnico, Universidade de Lisboa,* Lisbon, Portugal
email: forward.huang@gmail.com, db62560@connect.um.edu.mo, db62673@connect.um.edu.mo, cslam@um.edu.mo, pimak@um.edu.mo, rmartins@um.edu.mo

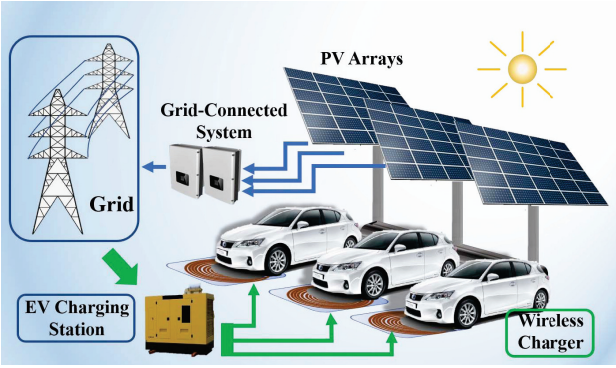
***Abstract*—In order to alleviate energy shortage and environmental pollution, there is an increasing demand to exploit renewable energy and promote electric vehicle (EV). It is relatively mature and easy to exploit widely distributed solar energy, which can be converted to clean electricity by using commercially available photovoltaic (PV) panels. In this paper, we propose a self-contained solar-powered wireless EV charging system, which operates independently without interaction with the power grid, resulting in low building cost and wide selection of building sites. Moreover, inductive power transfer (IPT) technology is incorporated in the proposed system for wireless charging, bring the advantages of friendly experience, safe operation and low maintenance cost. The proposed system has minimum converter stage, where the IPT converter is driven by the PV panels and loaded by the battery directly. Operation of maximum power point tracking (MPPT) for the PV panels and maximum efficiency point tracking (MPET) for the IPT converter are theoretically analyzed and experimentally verified.**

***Index Terms*—Inductive power transfer, photovoltaic, electric vehicle charging, maximum power point tracking, maximum efficiency point tracking**

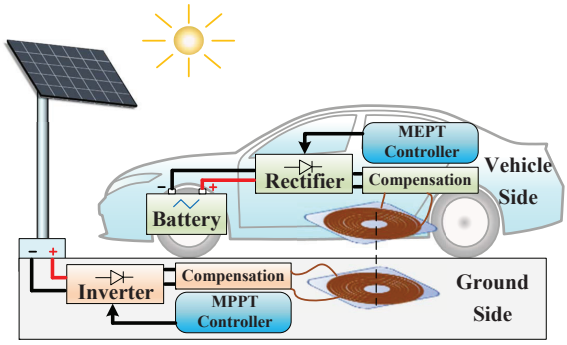
I. Introduction

Nowadays, energy shortage and environmental pollution are two major challenges to sustainable development of our economy. Petroleum, as it is widely known, with high energy density, is most suitable for our personal transportation. The number of petrol-vehicles worldwide is expected to be increasing in the near future, because of the rapid increase of purchasing power in developing countries, making it difficult to reduce the dependence on petroleum [1]. Meanwhile, to make matters worse, petrol-vehicles emit a large amount of harmful tailpipe pollutants and green gases, resulting in environmental problems in the cities [1]. Fortunately, with the development of electric vehicle (EV) technology, personal transportation powered by electricity rather than petroleum is a promising way to help addressing the problems of energy shortage and environmental pollution. In one hand, EVs are environment-friendly especially in the cities, because they are powered by clean electricity and have no exhausted emissions [2], [3]. In the other hand, clean electricity can come from widely-distributed renewable energies, i.e., solar power, biogas power, wind power and so on [4]. Nowadays, with the development of photovoltaic (PV) technology, it is a popular way to exploit solar power for EV charging [5].

This work was supported in part by the Science and Technology Development Fund, Macao SAR (FDCT) (025/2017/A1 and SKL-AMSV Fund) and in part by the Research Committee of University of Macau (MYRG2017-00090-AMSV, UM Macao Postdoctoral Fellowship).



(a) Centralized solar-powered EV charging station.



(b) Proposed self-contained solar-powered EV charging system.

Fig. 1. Two types of solar-powered EV charging systems.

Solutions of solar-power exploitation for EV charging can be identified as centralized and self-contained types. Fig. 1(a) shows a typical centralized solar-powered EV charging station, where the solar power is collected by PV panels and converted to electricity for EV charging. In order to balance the supply

power and the demand power, connecting to the power grid is usually necessary for the centralized solar-powered EV charging station, resulting in high building cost and limited building sites [4]. Alternatively, we propose a self-contained solar-powered EV charging system as shown in Fig. 1(b), where the power from the PV panels is solely responsible for a single EV. No additional converter is required to interact with the power grid or the battery for power balance or energy storage. Obviously, this simplest type of solar-powered EV charging system can cut down the building cost and suitable for modular design [4]. There is more freedom to select the building sites, because the self-contained solar-powered EV charging station can be built where the power grid is unavailable.

Conventional conductive charging has been commercially available, but it is not convenient for the users due to the plug-in and plug-out operations [6]. Apart from unfriendly user experience, it is not suitable for the exploitation of distributed renewable energy. Usually, renewable energy is distributed where the environment is hostile, e.g., high-temperature, moisture. If there are wear and tear losses in the electric cables, dangerous electric shock may happen and the maintenance cost will be high [6]. To address the above disadvantages of conductive charging, we are motivated to investigate wireless charging in the self-contained solar-powered EV charging station. Development in modern power electronics has enabled wireless inductive power transfer (IPT), making a user-friendly and maintenance-free interface possible in EV charging applications [7]–[9].

In the diagram of solar-powered wireless EV charging station shown in Fig. 1(b), the IPT converter is driven by PV panels directly and loaded by the onboard battery which is a voltage source. Such that, the output power of the PV panels is determined by the input resistance of the IPT converter, while the efficiency of the IPT converter is determined by its load resistance. Due to the rather complex relationship between the input and output characteristics of the IPT converter, it is a challenge to maintain maximum power point tracking (MPPT) for the PV panels and maximum efficiency point tracking (MPET) for the IPT converter. This paper proposes a self-contained solor-powered IPT system for wireless EV system. The system is modeled and analyzed to derive the optimum operations for MPPT and MPET. Experiment results validate the propose operation.

II. Modeling

A. System Structure

Fig. 2 shows the system structure of the proposed self-contained solar-powered wireless EV charging system, which includes PV panels, a series-series (SS) compensated IPT converter and an onboard battery. Inside the SS IPT converter, the magnetic coupler has self-inductance L_P and L_S, and mutual inductance M. Subscripts P and S indicate parameters in the ground side and the vehicle side, respectively. The coupling coefficient is given by $k = \frac{M}{\sqrt{L_P L_S}}$. C_P and C_S are the series compensation capacitors in the ground side and

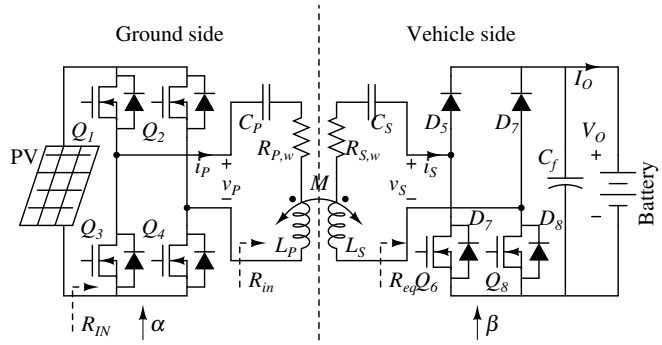


Fig. 2. System structure.

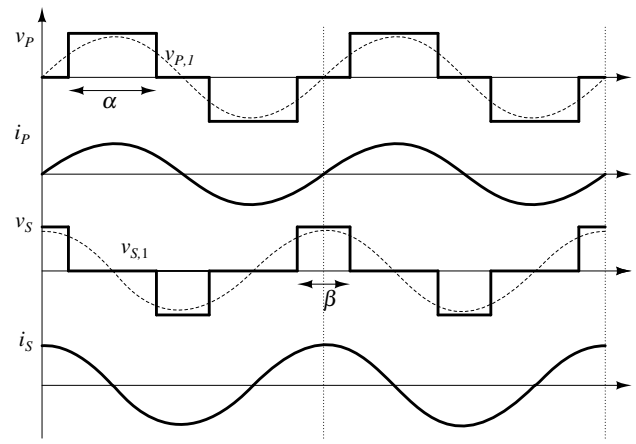


Fig. 3. Operating waveforms.

the vehicle side, respectively. Coil losses are represented by resistances R_P and R_S. The PV panels act as a voltage source and is modulated into a high-frequency ac voltage v_P to drive the resonant circuit in the ground side, by a full-bridge inverter having four SiC MOSFET switches $Q_1 - -Q_4$. The ac output is rectified to a dc output for battery charging by a semi-active rectifier. v_S and i_S are the terminal voltage and current of the semi-active rectifier, insisting of two diodes D_5, D_7 in the upper legs and two SiC MOSFET switches Q_6, Q_8 in the lower legs. C_P and C_S resonate with L_P and L_S at angular frequencies $\omega_P = \frac{1}{\sqrt{L_P C_P}}$ and $\omega_S = \frac{1}{\sqrt{L_S C_S}}$, respectively.

B. Operating Waveforms

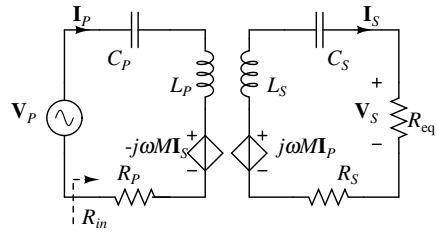


Fig. 4. Equivalent circuit model of the proposed system.

The operating waveforms of the IPT converter are shown in Fig. 3. Switches of the inverter and the semi-active rectifier are turned ON and OFF with appropriate driving sequence to generate the waveforms. The fundamental component of v_P is aligned with that of i_P, such that the input resistance can be transformed. Similarly, the fundamental component of v_S is aligned with that of i_S to modulate the load resistance. The conduction angles of v_P and v_S are α and β, respectively.

C. Equivalent Model

Fig. 4 shows the equivalent model of the proposed system using fundamental approximation. Since the IPT converter operates at resonant frequency and is a high-quality-factor circuit, this model is sufficiently accurate for subsequent analysis [10], [12], [13], [15]. Here, $\mathbf{V}_P$, $\mathbf{I}_P$, $\mathbf{V}_S$ and $\mathbf{I}_S$ are phasors of the fundamental components of v_P , i_P , v_S and i_S, respectively. Taking fundamental components into consideration, basic equations for the circuit model are

$$(R_P + jX_P)\mathbf{I_P} - jX_M\mathbf{I}_S = \mathbf{V}_P, \tag{1}$$

$$-(R_S + R_{eq} + jX_S)\mathbf{I}_S + jX_M\mathbf{I}_P = 0. \tag{2}$$

where

$$X_M = \omega M, \tag{3}$$

$$X_P = \omega L_P - \frac{1}{\omega C_P}, \text{and} \tag{4}$$

$$X_S = \omega L_S - \frac{1}{\omega C_S} \tag{5}$$

are the mutual reactance, the transmitter-side reactance and receiver-side reactance, respectively.

III. Analysis of Optimum Operation

A. Operation for MEPT

As the circuit model shown in Fig. 4, the efficiency can be highlighted by [14], [15]

$$\begin{aligned}\eta &= \frac{|\mathbf{I}_S|^2 R_{\text{eq}}}{|\mathbf{I}_S|^2 R_{\text{eq}} + |\mathbf{I}_S|^2 R_S + |\mathbf{I}_P|^2 R_P} \\ &= \frac{X_M^2 R_{\text{eq}}}{[(R_{\text{eq}}+R_S)^2+X_S^2]R_P+X_M^2(R_{\text{eq}}+R_S)}.\end{aligned} \tag{6}$$

With (6), the optimum operating frequency and the optimum load resistance for maximum efficiency are given by

$$\omega = \omega_S, \text{ and} \tag{7}$$

$$R_{\text{eq,opt}} = X_M\sqrt{\frac{R_S}{R_P}}, \tag{8}$$

Without loss of generality, we assume that maximum power can be extracted from the PV panels and the losses of the IPT converter can be neglected. Such that, the input power and the output power of the IPT converter are identical, given by

$$P_{OUT} = P_{IN} = P_{Max} \tag{9}$$

Thus, the battery voltage can be calculated as

$$R_L = \frac{V_O^2}{P_{Max}} \tag{10}$$

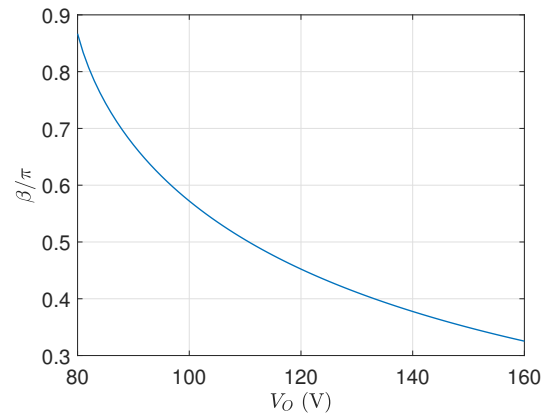


Fig. 5. Simulated conduction angle β versus battery voltage V_O for MEPT.

As the battery resistance should be transformed to an optimum value for the IPT converter, the conduction angle of the semi-active rectifier should satisfy

$$R_{eq,\text{opt}} = \frac{8}{\pi^2} R_L \sin^2\frac{\beta}{2} \tag{11}$$

and derived as

$$\beta = 2\arcsin\left(\sqrt{\frac{X_M\sqrt{\frac{R_S}{R_P}}}{\frac{8}{\pi^2}\frac{V_O^2}{P_{Max}}}}\right). \tag{12}$$

Equation (15) gives the optimum operation of the conduction angle β of the semi-active rectifier to maintain maximum efficiency by transforming the battery resistance. Obviously, β is determined by the maximum input power P_{Max}, the battery voltage V_O and some other parameters of the IPT converter. Fig. 5 shows the simulated β versus V_O with the following simulation parameters: P_{max}=120 W, $L_P = 254$ μH, $L_S = 364$ μH, $k = 0.25$, $C_P = 39.9$ nF, $C_S = 27.8$ nF.

B. Operation for MPPT

If the semi-active rectifier tightly transform the load resistance to an optimum value, with the circuit model shown in Fig. 4, the ac input resistance can be derived as

$$R_{in} = X_M\sqrt{\frac{R_P}{R_S}} \tag{13}$$

Typically, the output power of the PV panels is determined by the DC input resistance of the IPT converter. Supposed that maximum power can be extracted at $R_{IN,Max}$, the conduction angle α of the inverter should satisfy

$$R_{IN,\text{Max}} = \frac{R_{in}}{\frac{8}{\pi^2}\sin^2\frac{\alpha}{2}} \tag{14}$$

and derived as

$$\alpha = 2\arcsin\left(\sqrt{\frac{X_M\sqrt{\frac{R_P}{R_S}}}{\frac{8}{\pi^2}R_{IN,\text{Max}}}}\right). \tag{15}$$

TABLE I
CONVERTER PARAMETERS

Parameters	Symbols	Values
Input voltage	V_I	0–80 V
Battery voltage	V_O	80–160 V
MOSFETs	Q_1-Q_4,Q_6,Q_8	SCT3022AL
Diodes	D_5,D_7	FFSP1065A
Self inductance	L_P, L_S	255 μH, 364 μH
Compensation capacitance	C_P, C_S	39.8 nF, 27.8 nF
Coupling coefficient	k	0.255
Coil resistance	R_P, R_S	0.35 Ω, 0.95 Ω
Operating frequency	$\frac{\omega}{2\pi}$	50 kHz

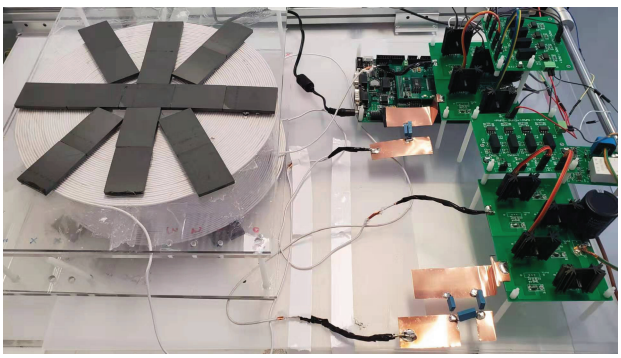


Fig. 6. Experimental prototype.

Equation (15) gives the optimum operation of the conduction angle α of the semi-active rectifier to extract maximum power by transforming the input resistance of the IPT converter. Obviously, α is determined by the illumination condition.

IV. EXPERIMENTAL VERIFICATION

To evaluate the proposed operation for MPPT and MPET in a self-contained wireless EV charging system, a down-scale experiment prototype is built as shown in Fig. 6 with parameters given in Table I. An electronic load is used to emulate the battery with terminal voltage ranging from 80 V to 160 V, while a programmable DC power supply is used to emulate the PV panels. The DC input power and DC output power are measured by a WT1800 power analyzer.

A. Measured Waveforms of MEPT

Experiment of MEPT is first conducted by setting the output power of PV panels constant and varying the conduction angle of the semi-active rectifier. The measured waveforms of the proposed system are shown in Fig. 7. It can be observed that, the input voltage v_P and input current i_P of the IPT converter are nearly identical for a constant input power from the PV panels. It can also be observed that, the waveforms of i_S are approximately sinusoidal and the magnitudes are nearly identical for different values of battery voltage. From the waveforms of v_S, the conduction angles β varies with the battery voltage V_O to maintain an optimum load resistance $R_{eq,\text{opt}}$. A direct read-out of the measured efficiency η and the conduction angle β versus battery voltage V_O is shown in

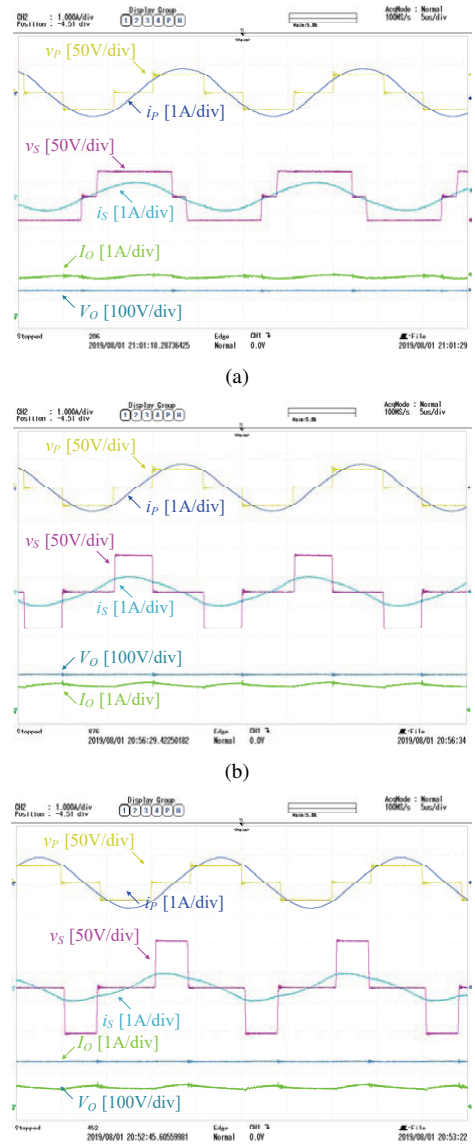


Fig. 7. Measured operating waveforms of the proposed system when the values of the battery voltage are (a) $V_O = 80$ V, (b) (a) $V_O = 120$ V and (c) (a) $V_O = 150$ V, given an identical maximum input power $P_{IN} = 119$ W.

Fig. 8, where MEPT is maintained under different status of the battery voltage.

B. Measured Waveforms of MPPT

Experiment of MPPT is then conducted under the condition that the semi-active rectifier maintained an optimum load resistance $R_{eq,\text{opt}}$ in regardless of the battery voltage. Fig. 7 shows the waveforms of the proposed system operating at different points of the PV output profile. From the waveforms of v_P and i_P, the input resistance of the IPT converter is changed by varying the conduction angle α of the inverter. A direct read-out of the measured output power of the PV panels versus input resistance of the IPT converter is shown in Fig. 8, where a maximum power of 119 W is extracted when the input

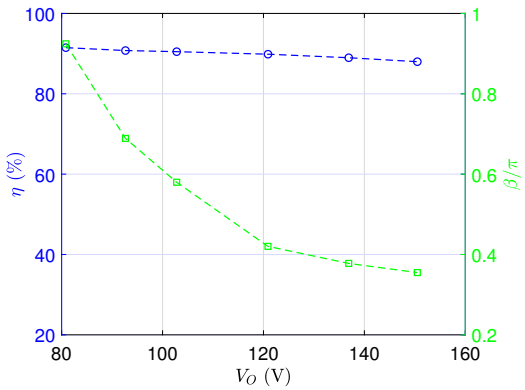


Fig. 8. Measured efficiency η versus battery voltage V_O.

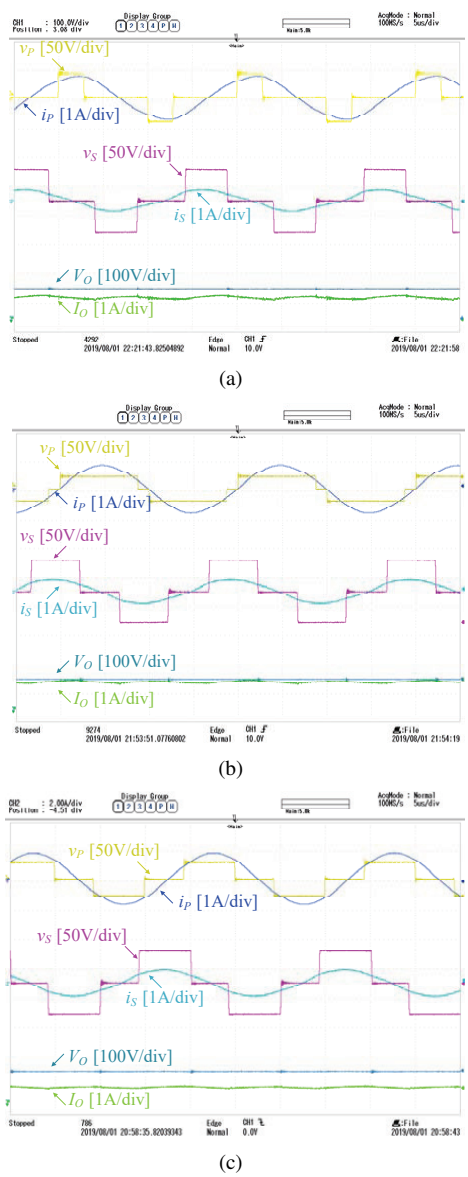


Fig. 9. Measured operating waveforms at (a) Point A, $P_{IN} = 107$ W, (b) Point B, $P_{IN} = 119$ W and (c) Point C, $P_{IN} = 88$ W, given an identical optimum load resistance.

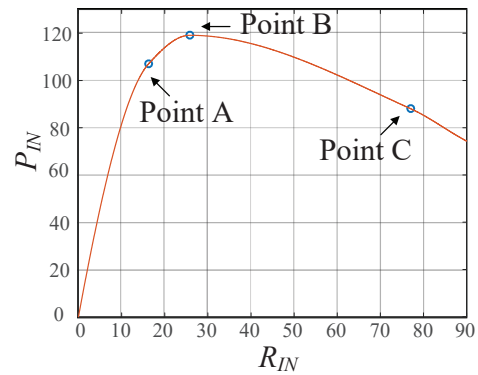


Fig. 10. Measured output power of the PV panels versus input resistance of the IPT converter.

resistance is set at 26 Ω, as Point B indicates. If not located at this optimum point, MPPT can not be achieved.

V. Conclusion

A self-contained solar-powered wireless electric vehicle (EV) charging system is proposed in this paper, with advantages of low building cost and free choice of building sites. An optimum operation approach for maximum power point tracking as well as maximum efficiency point tracking is theoretically analysis and experimentally verified. The proposed system as well as the optimum operation approach makes a simple and effective solution to exploit the solar power for EV charging. To make it perfect, we will study and implement close-loop control in the future work.

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